

CA2 Lab — Formula Sheet

Common setup

Frequency filtering pipeline: image $\rightarrow$ FFT $\rightarrow$ shift (centre) $\rightarrow$ multiply by H $\rightarrow$ inverse shift $\rightarrow$ inverse FFT $\rightarrow$ take real part $\rightarrow$ clip to $[0, 255]$

Distance grid (image $M \times N$, centre at $(M/2, N/2)$):

$$D(u, v) = \sqrt{(u - M/2)^2 + (v - N/2)^2}$$

D_0 = cut-off radius, n = Butterworth order, W = band width.

Spatial filtering: slide the kernel w over the image, output = sum of (kernel $\times$ pixels under it).

Image Enhancement — Spatial Domain

Box filter (low-pass)

$$m \times n \text{ kernel, every entry} = \frac{1}{mn} \quad \text{e.g.} \quad \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Gaussian filter (low-pass)

$$w(s, t) = e^{-\frac{s^2+t^2}{2\sigma^2}}, \quad \text{then divide by the sum of all entries}$$

s, t run from $-k$ to k (kernel size $2k+1$, typically $\approx 6\sigma$).

High-pass filter

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

General $N \times N$: all -1 , centre $= N^2 - 1$.

Unsharp masking

$$\text{mask} = f - \text{blur}(f), \quad g = f + \text{mask}$$

High-boost filter

$$g = f + k \cdot \text{mask}, \quad k > 1 \quad \text{or} \quad g = A f - \text{blur}(f), \quad A \geq 1$$

Kernel form: the 3×3 high-pass kernel with centre $x > 8$ (larger x keeps more of the original).

Image Enhancement — Frequency Domain

Low-pass masks

Ideal $H = 1$ if $D \leq D_0$, else 0

Butterworth $H = \frac{1}{1 + (D/D_0)^{2n}}$

Gaussian $H = e^{-D^2/(2D_0^2)}$

High-pass masks

$$H_{HP} = 1 - H_{LP}$$

Ideal $H = 0$ if $D \leq D_0$, else 1

Butterworth $H = \frac{1}{1 + (D_0/D)^{2n}}$ (set $H = 0$ where $D = 0$)

Gaussian $H = 1 - e^{-D^2/(2D_0^2)}$

High-frequency emphasis

$$H = k_1 + k_2 H_{HP}$$

$k_1 \approx 0.5$ (keeps low frequencies), $k_2 \approx 0.75-2$ (boosts high frequencies). H_{HP} usually Gaussian.

Ideal band-reject

$$H = 0 \text{ if } D_0 - \frac{W}{2} \leq D \leq D_0 + \frac{W}{2}, \quad \text{else } 1$$

Ideal band-pass

$$H = 1 \text{ if } D_0 - \frac{W}{2} \leq D \leq D_0 + \frac{W}{2}, \quad \text{else } 0 \quad (= 1 - H_{\text{reject}})$$

Or with two radii: $H = 1$ if $D_L \leq D \leq D_H$.

Homomorphic Filter

Steps: $z = \ln(1 + f) \rightarrow \text{FFT} + \text{shift} \rightarrow \text{multiply by } H \rightarrow \text{inverse shift} + \text{IFFT} \rightarrow g = e^{z'} - 1 \rightarrow \text{normalise to } [0, 255]$

Filter:

$$H = (\gamma_H - \gamma_L) \left[1 - e^{-c D^2 / D_0^2} \right] + \gamma_L$$

Typical: $\gamma_L = 0.5$ (< 1 , suppresses illumination), $\gamma_H = 2$ (> 1 , boosts detail), $c = 1$, $D_0 \approx 30$.

Noise (adding noise to image f)

$U = \text{uniform random in } [0, 1)$, clip result to $[0, 255]$.

Gaussian $g = f + n, \quad n \sim \mathcal{N}(\mu, \sigma^2) \quad (\text{e.g. } \mu = 0, \sigma = 20)$

Uniform $g = f + n, \quad n = a + (b - a)U \quad (\text{e.g. } a = -20, b = 20)$

Impulse (salt & pepper) for each pixel: if $U < p/2 \Rightarrow 0$; if $U > 1 - p/2 \Rightarrow 255$; else unchanged

Speckle $g = f + f \cdot n, \quad n \sim \mathcal{N}(0, \sigma^2) \quad (\text{e.g. } \sigma = 0.1-0.2)$

Poisson $g = \text{Poisson}(\lambda = f) \quad (\text{each pixel drawn with its own value as mean})$

Rayleigh $g = f + n, \quad n = a + \sqrt{-b \ln(1 - U)}$

Periodic $g = f + A \sin\left(2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N}\right)\right) \quad (\text{e.g. } A = 30)$

Denoising Filters

All operate on the $m \times n$ window S around each pixel.

Mean / averaging $\hat{f} = \frac{1}{mn} \sum_S g$

Weighted average $\hat{f} = \frac{\sum w g}{\sum w}, \quad w = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} \quad (\text{sum} = 16)$

Median $\hat{f} = \text{median of } S \quad (\text{best for salt \& pepper})$

Maximum $\hat{f} = \max(S) \quad (\text{removes pepper})$

Minimum $\hat{f} = \min(S) \quad (\text{removes salt})$

Alpha-trimmed mean sort S , drop $d/2$ smallest and $d/2$ largest: $\hat{f} = \frac{1}{mn - d} \sum (\text{remaining})$

Anisotropic diffusion (Perona-Malik)

Repeat for T iterations ($\lambda \leq 0.25$, $K = \text{edge threshold e.g. } 20-30$):

$$\nabla_N = I(x-1, y) - I, \quad \nabla_S = I(x+1, y) - I, \quad \nabla_E = I(x, y+1) - I, \quad \nabla_W = I(x, y-1) - I$$

$$c = e^{-(\nabla/K)^2} \quad \text{or} \quad c = \frac{1}{1 + (\nabla/K)^2}$$

$$I \leftarrow I + \lambda (c_N \nabla_N + c_S \nabla_S + c_E \nabla_E + c_W \nabla_W)$$

Fourier-based denoising

Random noise: multiply spectrum by a low-pass mask (Gaussian / Butterworth).

Periodic noise: find the bright spikes at $\pm(u_0, v_0)$ in $|F|$ and zero them out — band-reject at their radius, or a notch:

$$H = 0 \text{ if } D_1 \leq D_0 \text{ or } D_2 \leq D_0, \text{ else } 1, \quad D_{1,2} = \sqrt{(u - M/2 \mp u_0)^2 + (v - N/2 \mp v_0)^2}$$

Image Restoration

Blur model: $G = H \cdot F$ (+ noise). Blur an image by multiplying its spectrum by H , or convolving with PSF h .

Point spread functions

Out-of-focus disk: $h = 1$ if $x^2 + y^2 \leq R^2$, else 0; then divide by sum

Atmospheric turbulence $H = e^{-k[(u-M/2)^2 + (v-N/2)^2]^{5/6}}$ ($k = 0.0025$ severe, 0.001 mild, 0.00025 low)

Motion blur $H = \frac{T}{\pi(ua + vb)} \sin(\pi(ua + vb)) e^{-j\pi(ua + vb)}$ (use u, v centred; e.g. $a = b = 0.1$, $T = 1$; set $H = T$ where $ua + vb = 0$)

Spatial alternative: $1 \times L$ line of $1/L$, rotated to angle θ

Inverse filter

$$\hat{F} = \frac{G}{H}$$

Blows up where $H \approx 0$ — usually restrict to $D \leq D_0$ (set $\hat{F} = G$ or 0 outside).

Pseudo-inverse filter

$$\hat{F} = \frac{G}{H} \text{ if } |H| \geq \varepsilon, \quad \hat{F} = 0 \text{ otherwise} \quad (\varepsilon \approx 0.01\text{--}0.1)$$

Wiener filter

$$\hat{F} = \frac{H^*}{|H|^2 + K} G$$

H^* = complex conjugate of H ; K = noise-to-signal ratio (tune, e.g. 0.001–0.1). $K = 0$ gives the inverse filter.

Evaluation

$$\text{MSE} = \frac{1}{MN} \sum (f - \hat{f})^2, \quad \text{PSNR} = 10 \log_{10} \frac{255^2}{\text{MSE}} \text{ dB}$$